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INVOICE AGENT

Complete PDF and ZIP Web Application Guide

Setup, operate, secure, test, and troubleshoot the Next.js and FastAPI invoice checking application.

FRONTEND	BACKEND	BATCH SECURITY
Next.js 16 TypeScript Tailwind CSS	FastAPI Pydantic LangGraph	Path validation Size limits ZIP bomb controls

Version 1.1

Repository: souravhub/invoice-agent

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01 Purpose and architecture

Invoice Agent accepts one text-based invoice PDF or one ZIP containing up to 10 PDFs. It extracts structured invoice data, applies deterministic completeness and arithmetic checks, and returns either Passed configured checks or Needs review with reasons.

Decision boundary

A passing result is not payment authorization and is not proof that an invoice or vendor is authentic.

Technology layers

Layer	Technology	Responsibility
Browser	Next.js, TypeScript, Tailwind	Upload PDF/ZIP, show individual or batch results, download JSON
API	FastAPI, Pydantic	Upload limits, CORS, temporary files, documented responses
ZIP	Python zipfile	Archive validation, safe streaming, sequential batch control
Agent	LangGraph, LangChain	Structured extraction, retries, validation, review decision
PDF	pdfplumber	Parse embedded text; OCR is intentionally excluded

Request flow

```
Browser -> FastAPI -> isolated temporary directory
|
PDF -> process_invoice() -> LangGraph
ZIP -> zip_service -> each PDF sequentially -> process_invoice()
|
JSON response -> temporary directory deleted
```

02 Install and configure

Prerequisites

- Python 3.12 or newer.
- Node.js 20.9 or newer; Node.js 22 LTS is recommended.
- An OpenAI or Google Gemini API key.

Install from the repository root

```
cd D:\Trigital\invoice-agent
python -m venv venv
.\venv\Scripts\python.exe -m pip install --upgrade pip
.\venv\Scripts\python.exe -m pip install -r backend\requirements.txt
npm --prefix frontend install
```

Create configuration

```
Copy-Item backend\.env.example backend\.env
Copy-Item frontend\.env.example frontend\.env.local
```

backend/.env

```
AI_PROVIDER=openai
AI_MODEL=gpt-4o-mini
OPENAI_API_KEY=replace-with-your-real-key
GEMINI_API_KEY=replace-with-your-real-key
CORS_ORIGINS=http://localhost:3000
```

frontend/.env.local

```
NEXT_PUBLIC_API_BASE_URL=http://localhost:8000
```

Keep secrets backend-only

Real provider keys belong only in backend/.env. NEXT_PUBLIC values are visible in the browser.

03 Run and use the application

Terminal 1 - FastAPI

```
.\venv\Scripts\python.exe -m uvicorn backend.main:app --reload --host 127.0.0.1 --port 8000
```

Terminal 2 - Next.js

```
npm --prefix frontend run dev
```

URLs

URL	Purpose	Expected result
http://localhost:3000	Web app	PDF or ZIP upload screen
http://localhost:8000/api/health	Health	{"status":"ok"}
http://localhost:8000/docs	OpenAPI	Interactive endpoint documentation

Single PDF

- Choose one text-based PDF up to 20 MB and click Check invoice.
- Review fields, line items, amounts, status, attempts, and reasons.
- Download the complete result as JSON.

ZIP batch

- Choose one ZIP up to 20 MB compressed containing 1 to 10 PDFs.
- Click Check batch; PDFs are processed sequentially.
- Review the batch summary and expand each invoice card.
- A bad or scanned PDF has its own error while successful results remain available.
- Download the complete batch JSON.

Keep both terminals open

If the browser cannot reach the invoice service, open the health URL first and restart FastAPI if it is unavailable.

04 API contract

POST /api/invoices/check

Multipart field file. Accepts one application/pdf up to 20 MB and returns invoice, status, reasons, and attempts.

POST /api/invoices/check-batch

Multipart field file. Accepts one ZIP up to 20 MB compressed. An accepted archive returns one file record per PDF and HTTP 200 even when individual PDFs fail.


```
{
"summary": {"total": 2, "succeeded": 1, "failed": 1},
"files": [
{
"filename": "invoice_01.pdf",
"result": {"invoice": {}, "status": "auto_approved", "reasons": [], "attempts": 1},
"error": null
},
{
"filename": "scan.pdf",
"result": null,
"error": {"code": "unreadable_pdf", "message": "No readable text was found."}
}
]
}
```


Batch limits

Limit	Value	Purpose
Compressed upload	20 MB	Bounds request storage and transfer
PDF count	10	Bounds AI calls and response duration
One PDF uncompressed	20 MB	Bounds per-entry memory/disk use
Total uncompressed	100 MB	Bounds decompression output
Compression ratio	100:1	Detects suspicious ZIP bomb behavior

05 ZIP processing and security

Implementation

- FastAPI streams the compressed upload into a unique TemporaryDirectory.
- zip_service.py inspects every ZipInfo before decompressing content.
- Each PDF is streamed to a generated UUID filename; archive paths are never used as disk paths.
- Actual decompressed bytes are counted while copying, not just trusted from ZIP metadata.
- Each generated PDF path is passed to the existing process_invoice() function.
- The entire batch runs in FastAPI's worker thread and is deleted afterward.

Rejected archives

Threat or invalid input	Backend response
Absolute, drive-letter, or .. traversal path	400 invalid_zip
Symbolic link or encrypted entry	400 invalid_zip
Nested ZIP or non-PDF regular file	400 invalid_zip
Too many PDFs or excessive expanded bytes	400 invalid_zip
Suspicious compression ratio	400 invalid_zip
Corrupt or unsupported archive compression	400 invalid_zip
Compressed upload over 20 MB	413 file_too_large

No extractall

The implementation never calls ZipFile.extract() or extractall(). This removes the archive entry name from filesystem path selection.

06 Agent rules and safe errors

Preserved workflow

```
PDF text -> AI extraction -> deterministic validation -> decision
^ |
`-- retry missing extraction data only
```

Rule	Outcome when triggered
Vendor, invoice date, or total missing	Needs review
No line items	Needs review
Currency missing	Needs review
Items - discount != subtotal within 0.01	Needs review
Subtotal + tax + shipping != total within 0.01	Needs review
More than one extraction attempt	Needs review

Per-file batch errors

Code	Meaning
invalid_pdf	Entry has a PDF name but is corrupt or cannot be parsed
unreadable_pdf	PDF contains no extractable text and needs OCR
ai_provider_error	Provider failed; sensitive provider details are hidden

Business controls still apply

The system does not verify vendor identity, detect fraud or duplicates, confirm receipt of goods, or authorize payment.

07 Testing and troubleshooting

Automated verification

```
.\venv\Scripts\python.exe -m pytest backend\tests -q
npm --prefix frontend run typecheck
npm --prefix frontend run build
```

ZIP test coverage

- Documented batch response and cleanup after archive failure.
- Partial success with a per-file unreadable PDF.
- Traversal, absolute, and Windows drive paths.
- Symbolic links, nested ZIPs, and non-PDF entries.
- Excessive PDF counts and suspicious compression ratios.
- AI processing is mocked, so tests spend no provider quota.

Troubleshooting

Symptom	Action
Could not reach invoice service	Open /api/health; start FastAPI and keep its terminal open.
Configuration error	Check provider, model, and selected provider key in backend/.env; restart FastAPI.
Invalid ZIP	Remove non-PDF/nested/encrypted/unsafe entries and confirm batch limits.
Unreadable PDF	Use a text-searchable PDF. OCR is outside this version.
CORS error	Match CORS_ORIGINS to the exact frontend origin and restart FastAPI.
Port in use	Stop the other process or update both the backend port and frontend API URL.

Ready state

Both servers stay running, /api/health returns ok, PDF and ZIP uploads produce structured results, and changing the upload clears old results.